HPD-RVE — Supplementary Material

S1. S24/48 — 24- vs 48-slip-system native cI comparison

Table 1: Direct comparison of the 24-slip-system cI (earlier simplification) versus the physically complete 48-slip-system cI (this work) as the native description, isolating the source of the cF/cI held-out ratio flip. Body-centred-cubic slip is experimentally known to operate on three plane families—{110}, {112}, and {123}—all with ⟨111⟩ Burgers vector, giving 48 distinct slip systems (Christian, 1983); the {123}⟨111⟩ family is required for *full* multi-slip accommodability (the five independent slip systems needed for an arbitrary shape change under the Taylor/Bishop–Hill condition). The 24-system simplification retains only {110}⟨111⟩ + {112}⟨111⟩ and therefore under-determines the slip subspace, leaving the polycrystal over-constrained and artificially elevating the Taylor factor (Table 1); this is why the prior 24-system reference yields the larger 1.76× cF/cI RMSE ratio—a slip-system-truncation artifact, not a defect of the cF proxy. The cF proxy (FCC, 12 systems, $M \approx 3.06$) is identical in both cases; only the native BCC slip-system set changes. The complete 48-system set is the standard BCC description in modern crystal-plasticity FE (e.g. DAMASK; Roters et al., 2019) and is adopted here. The FCC $M \approx 3.06$ is the standard uniaxial Taylor factor and matches the lattice-proxy analysis in the main text, where the small FCC/BCC Taylor-factor difference (BCC 48-system $M \approx 2.75$, ranging up to $M \approx 3.067$ in the literature) is shown to be absorbed by calibration.

Native cI set	BCC slip families	Effective Taylor factor M	Held-out cF/cI RMSE ratio
24-system (prior version)	{110}⟨111⟩ + {112}⟨111⟩	higher (fewer slip options ⇒ more constrained)	1.76 (native cI closer to HV)
48-system (this work)	+ {123}⟨111⟩	lower (≈ 2.75)	0.89 (cF marginally closer)

S2. Risk register

Table 2: Validation limitations and risk register. The three open concerns span independent validation (RWTH batch, copper fit, z-score leakage), microstructure representativeness (no EBSD cross-check, data not public), and interpretation of the BCC cF-proxy result flip.

Aspect	Status	Detail / required action
Independent validation	[!] **Partially addressed**	The 7-train/3-validation split reuses the same RWTH DP800 batch family; the pure-copper (FCC) case was never fitted to its own experiment; no input-feature normalization is applied, so there is no batch-statistic leakage from feature scaling—a within-batch consistency check, not a fully external held-out. A genuinely external second source is now available: POSTECH DP780 (Zenodo 10.5281/zenodo.5577981, CC BY 4.0, different lab and country). The RWTH-trained CP prediction reproduces the independent-lab DP780 flow curve with RMSE 114 MPa – comparable to the cross-lab experimental scatter (109 MPa) – so the model error on an external laboratory is no larger than lab-to-lab variability. Remaining gap: DP780 vs DP800 grade and phase-fraction/texture differences; an RVE-level (microstructure) external held-out still requires a second-source voxel/EBSD dataset.

Aspect	Status	Detail / required action
Microstructure representativeness	**[!] Addressed (aggregate); ODF benchmarked vs. Ashrafi 2023**	RVE phase fraction ($v_m \approx 0.28$) and near-unity phase-size ratio follow Tasan et al. (2014); an EBSD microtexture study of DP800 (Ashrafi & Aghili 2023, a different batch / heat-treatment reporting $\sim 48\%$ martensite and a yield strength of 485–489 MPa that is nearly direction-independent) is cited here only for texture weakness and yield isotropy, not for phase fraction; an independent commercial DP800 characterization (Aghadavoudi Jolfaei et al. 2021) reports ferrite grains of 2–6 μm and hard-phase fractions of $\approx$ 35–50% across heats, confirming the ferrite+martensite duality and sub-10 μm ferrite scale are representative of real DP800. The weak texture of DP steels is confirmed by EBSD on DP800 (Ashrafi & Aghili 2023: peak $\sim$ 2.5–2.6×random, yield direction-independent) and on DP780 (Ren et al. 2025; Badkoobeh et al. 2022), justifying the uniform-random prior. ODF benchmark: the $N_g = 32$ RVE orientations give a kernel-density ODF of mean 0.91×random with max 3.6×random at the 51st pct of the random null; the real DP800 EBSD peak ($\sim$ 2.6×random) lies within this sampling scatter, so the prior adds no spurious texture. Remaining gap: a point-by-point ODF overlay vs. a released EBSD map is future work (RWTH/Tasan raw EBSD not public).

Aspect	Status	Detail / required action
BCC cF proxy	**[!] Partially addressed**	The cF proxy is validated against the native-BCC full-field solve under uniaxial tension (gap 0.35%, RMSE 2.3 MPa; see the uniaxial strict held-out in the main text) and against two independent same-grade DP800 sources in leave-publication-out (RMSE 22–60 MPa). The 24- vs 48-system sensitivity of the native-BCC reference (RMSE ratio 1.76× → 0.89× on upgrade to the complete 48-system cI) is documented in Table 1; it reflects native-BCC reference richness, not a proxy defect. The uniaxial cF/cI comparison on the real reconstructed RWTH RVEs remains future work. Action: present the proxy as a fit-for-purpose surrogate.

S3. Priority-based parameter-resolution algorithm

The priority-based resolution engine fuses heterogeneous sources into solver-ready inputs and emits a machine-readable provenance record. It proceeds in four steps.

Input: experimental set D_{exp}, user set D_{user}, library set D_{pre}, inferred set D_{inf}; conflict threshold $\theta = 0.20$.

1. **Source-hierarchy fusion.** Assemble the fused set by strict priority experimental > user > library > inferred:

$$D_{\text{final}} = D_{\text{exp}} \cup \big(D_{\text{user}} \setminus D_{\text{exp}}\big) \cup \big(D_{\text{pre}} \setminus (D_{\text{exp}} \cup D_{\text{user}})\big) \cup \big(D_{\text{inf}} \setminus (D_{\text{exp}} \cup D_{\text{user}} \cup D_{\text{pre}})\big). \quad (1)$$

2. **Conflict detection.** For every parameter present in ≥ 2 sources, compute the relative deviation $\delta = |v_a - v_b| / \max(v_a, v_b)$. If $\delta > \theta$, retain the higher-priority value, flag a conflict, and archive the lower-priority alternative in the provenance log.

3. **Confidence tiering.** Assign each resolved parameter an ordinal tier: HIGH if sourced from experiment; MEDIUM if from user/library with no conflict; LOW if inferred/fallback or conflict-arbitrated.

4. **Downstream control.** Map the tier to computational actions: ensemble floor $N_{\text{floor}} \in \{3, 4, 5\}$ and UQ perturbation $\in \{\pm 2\%, \pm 5\%, \pm 20\%\}$ for (HIGH,MEDIUM,LOW); the nominal realization count is

$$N_i = \min\big(\max(\lceil M(C_i) \rceil, N_{\min}), N_{\max}\big), \tag{2}$$

$$M(C_i) = \alpha\, n_{\text{phase}} + \beta\, n_{\text{lattice}} + \gamma\, \text{CV}_d, \tag{3}$$

with α=1.0, β=0.5, γ=0.3, $N_{\min}$=1, $N_{\max}$=10; effective ensemble $N = \max(N_i, N_{\text{floor}})$.

Output: solver-ready parameter file and `dp800_fusion_provenance.json` recording source, tier, and resolution status for every parameter.